Secrets of Optical Flow- Report

Note: The output images have been stored in the `results` folder.

2.3 Analyzing Lucas Kanade Method

Ans 1:

If the structure tensor $A^T A$ has a rank of 2 it means it has 2 non zero eigenvalues corresponding to the directions of the principle image gradients. Such a point is the corner point which is significant from the point of view of determining optical flow. For smooth regions both eigenvalues are close to zero as the optical flow equation for all the pixels in the window turns out to be the same and therefore, our matrix $A^T A$ is not well conditioned and calculating optical flow becomes unreliable. Similarly in case of edge regions when one eigenvalue is significantly bigger and the other is close to zero, we get prominent gradient only in 1 direction similar to the problem we encounter in aperture problem. This again makes computing optical flow unreliable.

The role of threshold τ is that it helps us in determining if $A^T A$ has a rank of 2 or not. It is set to a small value like 0.01 and if the smaller eigenvalue is less than it, we discard that point for optical flow calculation as computing optical flow becomes unreliable in such cases. As explained above such regions are likely to be smooth regions or edge regions which are not well textured.

Ans 2:

The single scale Lukas Kanade algorithm implemented in this task works well for Grove3 and RubberWhale datasets. However, the results are not so satisfactory in Urban2 dataset. This can be seen from the Average End Point errors as well which is far more for Urban2 images than for RubberWhale or Grove3 indicating that the predicated flow is significantly different from ground truth flow in this case.

A reason for this could be that images in Urban2 have regions that have relatively poor texture so the optical flow calculated is less reliable due to absence of corner points (well textured regions) which are ideal for computing optical flow.

Further, it was observed in case of ground truth values of RubberWhale dataset that a very small number of the flow values were absurdly large and as a result the Average EPE was running into millions. So a threshold of 100 was set for ground truth flow values to remove these outliers while calculating the error and get a reasonable error at the end which makes sense while comparing with OpenCV's implementation of Lucas-Kanade algorithm.

Ans 3:

Upon experimenting with different window sizes, it was observed that error initially decreases from going from window size 8 to 10 and then starts increasing. So window size 10 gives us the best results in Lucas Kanade algorithm as it gives the least Average End Point Error.

	Window Size	Grove3	RubberWhale	Urban2
1	8	3.542527	0.988953	14.489106
2	10	3.542056	0.989338	14.485553
3	12	3.543193	0.991568	14.490197
4	16	3.546174	0.993805	14.500810
5	20	3.546116	1.000373	14.503330

In Lucas Kanade algorithm it is assumed that the optical flow for all the pixels in the window is the same. However, if window size becomes too big, this assumption doesn't hold good and our estimation of optical flow becomes less accurate and the AEPE increases.

At the same time if our window size is too small, we will get less number of equations to solve for optical flow ($Au = b$). So when we use least squares solution to find the solution for optical flow, our solution might not be the best i.e. it might not give us the least error. For example it might happen that in the small window there might be some pixels for which the motion is significant(typically for objects that are closer to us in the image) and thus higher optical flow and some pixels for which the motion is very little and thus there might be a lot of variation in the pixels of a small window. Due to this lack of spatial information we can end up getting higher errors in our optical flow estimation.

So an ideal window size is the one that is small enough to ensure that optical flow remains same for all the pixels in the window but is large enough to capture sufficient spatial information so that we can get the most accurate value of optical flow while using the least squares technique.

Ans 4:

HSV stands for hue saturation value. It is preferred to use HSV color space in ground truth visualisations because it is more intuitive to represent direction and magnitude of motion vectors this way. Hue represents the direction of motion vector and saturation its magnitude. So if the hue is uniform it indicates that direction of motion of motion vectors is not changing and similarly if hue varies, it indicates change in direction. For saturation, a high saturation values is indicative of large magnitude of motion and similarly a small value indicates small magnitude of motion. Such a visualisation is advantageous from a human perspective because it is highly intuitive to analyse.

3 Multi Scale Coarse to fine Optical flow

The average end point error for all 3 datasets of images is as follows

```
AEPE for Grove3using Multi Scale Lucas Kanade for window size 10 is:
3.24186199527771
AEPE for RubberWhaleusing Multi Scale Lucas Kanade for window size 10 is:
0.898669430994599
AEPE for Urban2using Multi Scale Lucas Kanade for window size 10 is:
14.089370071352317
```

As can be seen from the below table, the average end point error is relatively lower in case of Multi Scale Lucas Kanade algorithm than the single scale algorithm.

Image Dataset	Single Scale LK(AEPE)	Multi Scale LK(AEPE)
Grove3	3.54	3.24
RubberWhale	0.98	0.89
Urban2	14.48	14.09

This is because MSLK takes into account structure of image at different scales and so it can handle larger displacements better. for example it often happens that objects in the from tend to get displaced more than the objects behind in in the background. So applying Single scale LK algorithm is problematic for the objects in the front as their displacement is not small which is the fundamental assumption of that method. To overcome this, the image is scaled down so that even for objects in the front the displacement becomes small and optical flow can be computed just like for single scale implementation. Further this optical flow is upscaled to warp the upscaled version of the image and once again the optical flow can be computed and so on.

Thus, we observe that Multi Scale LK performs better than Single Scale LK for all the 3 given datasets.